

NITIK JAIN

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EDUCATION

(*EXPECTED)

Johns Hopkins University

Aug 2025 - May 2027*

MSE Robotics (Perception & Cognitive Systems)

- **Student Researcher, Johns Hopkins Medicine:** Deep Learning & AI for MRI enhancement and segmentation
- **Coursework:** Computer Vision, Deep Learning, Algorithms for Sensor-Based Robotics, Generative Models for Vision

Indian Institute of Technology (IIT), Kanpur

Jul 2017 - May 2022

Integrated B.Tech + M.Tech, Mechanical & Aerospace Engineering; GPA: 9.4/10, Minor in Machine Learning

- **Master's Thesis:** Object Detection Underneath Sparse Occlusion of Forests *Advisor: Dr. Mangal Kothari*
- **Academic Excellence Award 2021** — best academic performance amongst all students in the dual-degree program

WORK EXPERIENCE

Lead AI Engineer — Swiffylabs.ai, Bangalore, India

Jun 2024 - Jun 2025

WestBridge Capital-backed B2B fintech redefining core banking through applied AI; led AI development from 0→1

- Designed and deployed an AI-driven core banking stack, replacing legacy systems with **modular microservices**
- Integrated **LLM agents** and fraud-detection models into transaction, onboarding, and customer-experience workflows
- Partnered with NBFC clients to launch a **regulatory-ready, headless product suite**, leading the first commercial rollout

Founding AI Engineer — Osfin.ai, Bangalore, India

Apr 2023 - Jun 2024

Sequoia-backed fintech building AI-first document intelligence; led product vision for an LLM-powered accounting suite

- Led cross-functional teams to define and ship strategic product roadmaps for **LLM-powered** document analysis
- Built **OCR-based** invoice extraction in Bahasa Indonesia, improving extraction accuracy by **40%**
- Engineered **ETL** pipelines into a **Vector DB** with clustering, reducing retrieval search space by **80%**

AI Scientist, R&D Division — Qure.ai, Bangalore, India

Sept 2022 – Mar 2023

Deep Learning researcher for qCT-Chest — Qure.ai's flagship AI suite for early lung-cancer detection from CT scans

- Developed **3D ViT** and attention-enhanced **U-Net** models in **MONAI**, gaining **9% AUC** and **14% DICE** over SoTA
- Built an end-to-end **YAML-driven** training framework on **PyTorch Lightning + Hydra**, improving reproducibility
- Engineered scalable **Databricks** pipelines and automated annotation workflows, cutting data-prep and training time

AI Research Consultant — MooVita Pte Ltd, Singapore

Sept 2022 – Feb 2023

Perception researcher in Computer Vision, focusing on explainability and visual perception for autonomous driving

- Engineered a novel fallback system using **classical optical flow** for rapid vehicle-turn estimation at **240 FPS** on CPU
- Devised an **object-detection fallback pipeline** using **RAFT's** dense optical flow, exploiting motion-based velocity cues

KEY PROJECTS

Autonomous Ground Vehicle (AGV): Daksh II — IIT Kanpur 📺

Aug 2018 – May 2021

Built an autonomous ground vehicle for outdoor navigation; led the team at **IGVC 2019**, winning 4 international awards

- Developed an end-to-end **autonomy stack** in ROS spanning perception, SLAM, and motion planning for outdoor navigation
- Implemented low-latency **Fast-SCNN segmentation** with LiDAR-camera fusion for real-time lane detection and mapping
- Designed an **RRT-based trajectory planner** with cubic-spline smoothing for efficient path planning in ROS

Autonomous UAV for 3D Mapping & Path Planning — RRC, IIIT Hyderabad

May 2019 – Jul 2019

End-to-end ROS stack for mapping and 3D path planning under Prof. K. Madhava Krishna, IIIT Hyderabad

- Implemented **VINS-Mono** with IMU and **GPS-RTK** telemetry for accurate multi-sensor localization on ROS
- Built a **voxel-based 3D mapper** from RealSense data, enabling real-time trajectory generation via `mav_voxblox_planning`

ACADEMIC PUBLICATIONS & PATENT

- **N. Jain**, P. Yadav, R. Das, I. Saha, "Computer-Implemented Stigma Separation System" *Indian Patent No. 564565 (2025)*
- **N. Jain**, M. Shukla, S. Gupta, "IR Models and the COVID-19 Pandemic: A Comparative Study" *Europe PMC, 2023* 🔗
- **N. Jain**, A. Shah, H. B. Reddy, M. Kothari, "Design of a Low-Cost Autonomous Ground Vehicle" *IEEE ICARSC 2022* 🔗
- **N. Jain**, A. Sharma, M. Kothari, "Lightweight Multi-Drone Detection and 3D-Localization via YOLO" *ArXiv 2022* 🔗

SCHOLASTIC ACHIEVEMENTS

- **J. N. Tata Scholar 2025** — one amongst **100 scholars worldwide** selected for postgraduate studies abroad (2025)
- **K. C. Mahindra Scholar 2025** — one amongst **90 scholars worldwide** chosen for postgraduate education overseas (2025)
- **Lescoc Cup — 2nd Worldwide & 3rd AutoNav**, 27th Intelligent Ground Vehicle Competition, Michigan, USA (2019)

TECHNICAL SKILLS

Languages: Python (*Proficient*), C/C++, MATLAB, SQL, Bash, R

ML / DL: PyTorch, PyTorch Lightning + Hydra, MONAI, Hugging Face, LangChain + LangSmith, Scikit-learn, OpenCV

Robotics & Tools: ROS + Gazebo, Docker, Databricks, Grafana, CloudWatch, Git, Autodesk Inventor, $\LaTeX$